PROJECT PRESENTATION

AI Research Paper Skeptic Agent

A multi-agent RAG system that reads a research paper and produces a structured, evidence-checked skeptical review.

TRACK	DOMAIN	DIFFICULTY
AAI	Academic research support	Basic + Intermediate
DURATION	**MODE**	**REPOSITORY**
10 Days	Individual or Group of 3–4	Scholar_Research_Agent

Team: [Add team member names]

The Problem

Papers get summarized. They rarely get questioned.

Students and early-career researchers often restate a paper's claims without checking whether the paper's own evidence actually supports them. This project builds an agent that reads a paper and produces a skeptical review — surfacing the main claim, method, supporting evidence, limitations, and open questions a careful reader should ask.

Real-World Impact

A practical assistant for a real user need, keeping a human in the loop for judgment calls — the agent flags whether a claim is well-, weakly-, or unsupported by retrieved evidence rather than deciding accept/reject itself.

Who this is for

- **Students**

  doing literature reviews and journal clubs

- **Early-career researchers**

  screening papers before deep reading

- **Reviewers**

  wanting a first-pass grounding check

Dataset & Reference Source

Source

arXiv API and PDF documents (info.arxiv.org/help/api)

How it's used

Papers fetched by arXiv ID/URL or upload; text extracted, chunked, embedded, indexed for evidence lookup

Synthetic fallback

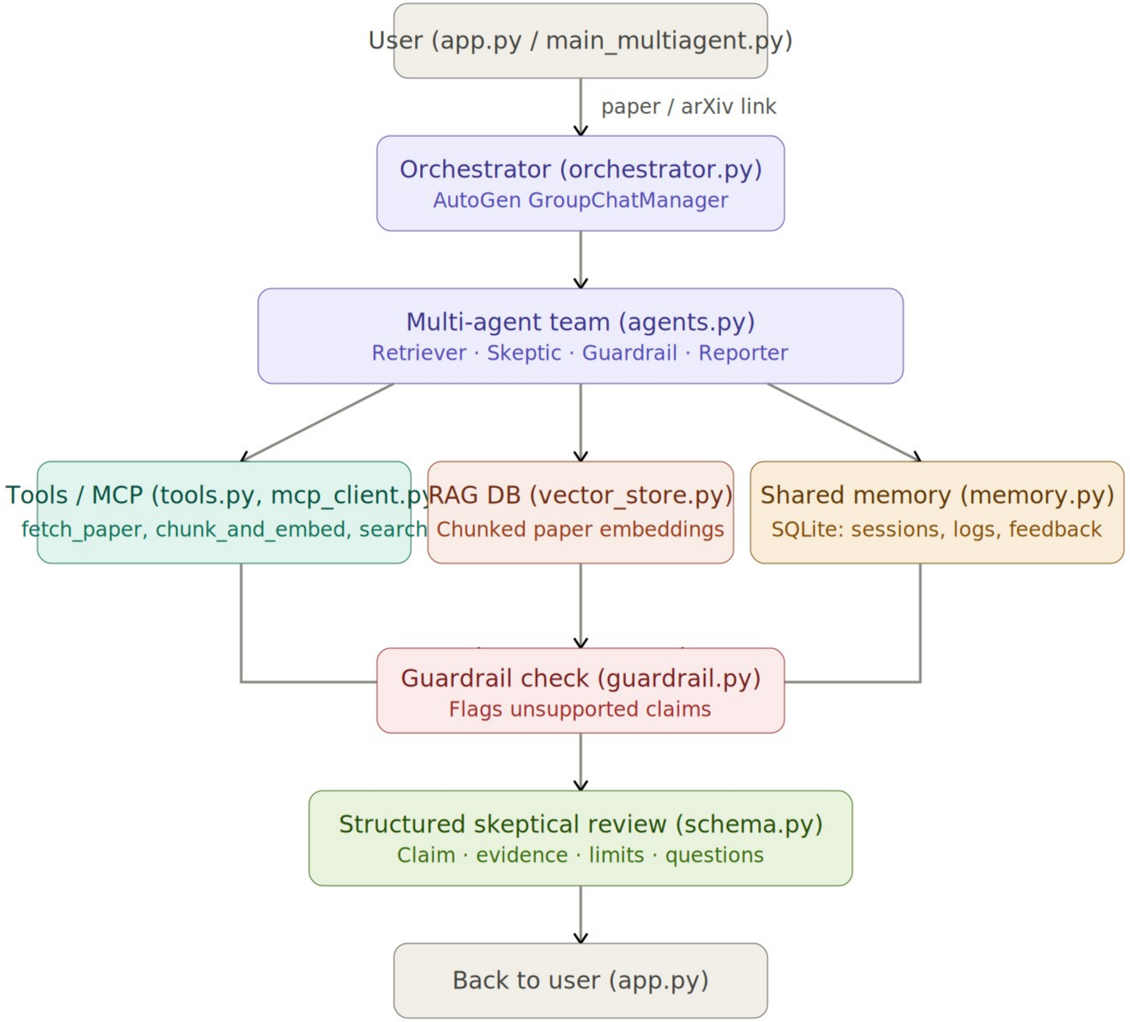
generate_sample_papers.py creates clearly-labeled synthetic sample text for testing when live access isn't available

Output target

Structured review (claim, method, evidence, limitations, questions) + a per-claim groundedness flag

System Workflow

User → Orchestrator → Multi-agent team → Guardrail → Structured review



Flow, step by step

1. User submits a paper or arXiv link
2. Orchestrator coordinates the AutoGen agent team
3. Retriever, Skeptic, Guardrail, Reporter collaborate via shared tools & memory
4. Every claim passes a grounding check before reaching the final review
5. Structured review + logs returned to the user

AI / Agent Innovation

Four mechanisms, each targeting a distinct failure mode of naive summarization

PDF RAG

Chunk, embed, and index the paper so claims are checked against its actual text — not the model's memorized understanding.

Multi-agent claim–evidence mapping

Retriever gathers passages; Skeptic drafts claims; Guardrail verifies grounding; Reporter assembles the review — generation is separated from verification.

Limitation extraction

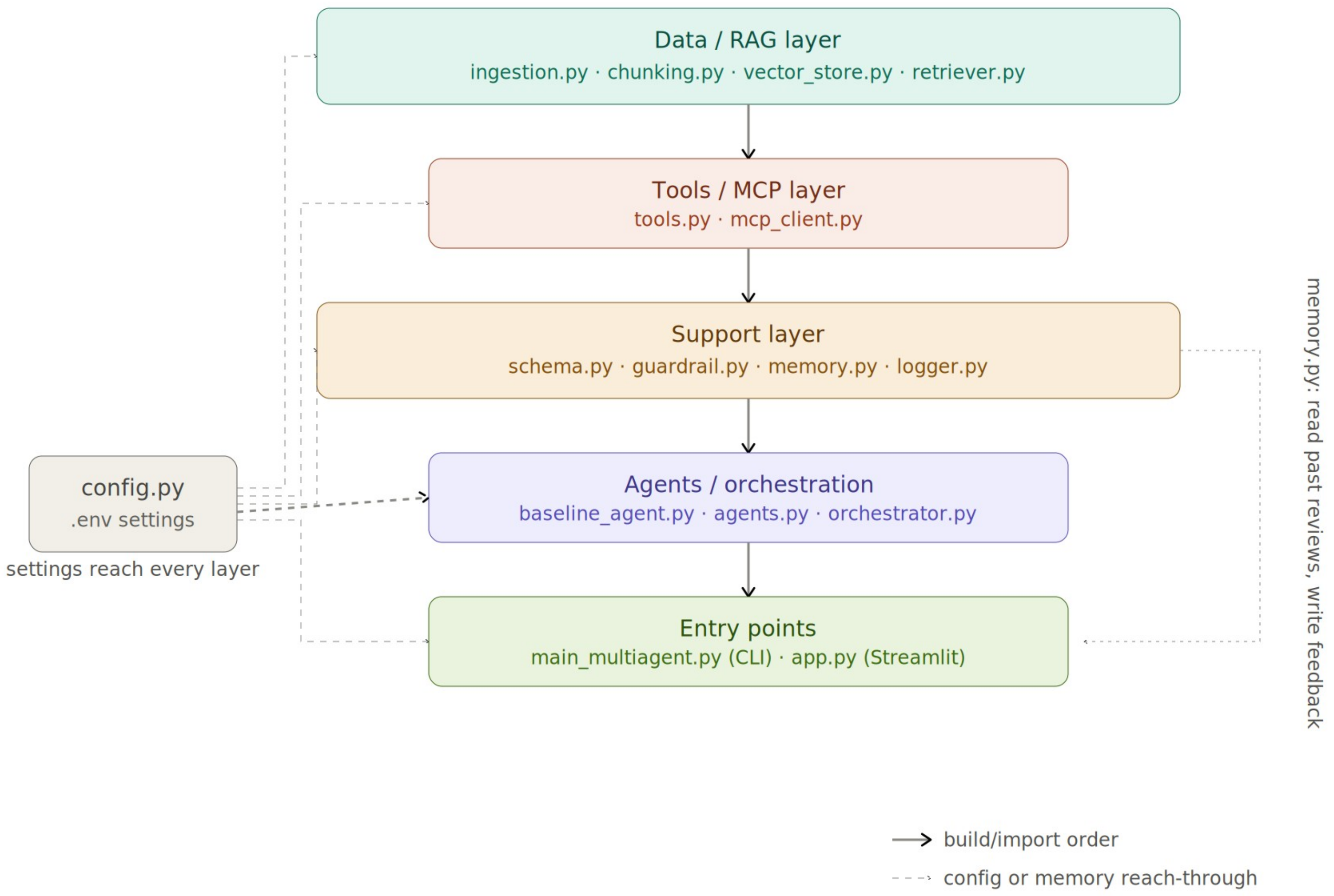
The Skeptic agent is prompted to surface method weaknesses, missing baselines, small samples, and unaddressed confounders.

Unsupported-claim guardrail

check_grounding() rejects/flags claims whose similarity to cited evidence falls below a threshold (default 0.6) — a quantitative, adjustable check.

Prototype Architecture

Six build layers — config.py reaches every layer; memory.py feeds session state back to the entry points



Layers

Data / RAG

ingestion · chunking · vector_store · retriever

Tools / MCP

tools.py · mcp_client.py

Support

schema · guardrail · memory · logger

Agents

baseline_agent · agents · orchestrator

Entry points

main_multiagent.py (CLI) · app.py (Streamlit)

Sample Output

Illustrative mock-up of a structured skeptical review (not an actual screen capture)

Review: “Scaling Sparse Attention for Long-Context Models”

MAIN CLAIM	Sparse attention matches dense attention quality at 4x lower compute on sequences > 32k tokens.
METHOD	Learned routing mask + block-sparse kernels, evaluated on 3 long-context benchmarks.
EVIDENCE	Retrieved passages: Table 3, Section 4.2 — benchmark scores and ablation results.
LIMITATIONS	No comparison against the strongest recent dense baseline; single random seed reported.
QUESTIONS TO ASK	Does the result hold beyond 64k tokens? Was the baseline tuned equally?

Groundedness Flags

Main claim
Well supported

Method
Well supported

Evidence scope
Partially supported

Limitations
Flagged — below threshold

Limitations & Responsible Use

Similarity ≠ entailment

The grounding check is cosine similarity, not logical entailment — topically similar but non-supporting passages can slip through.

Retrieval-dependent

Poorly segmented PDFs (multi-column, scanned/unOCR'd) degrade both retrieval and grounding accuracy.

A screening aid, not a verdict

Flagged claims still need human verification; this is not a substitute for peer review or expert judgment.

Not for high-stakes use alone

Should not be the sole basis for clinical, legal, or policy conclusions without expert review.

Future Improvements

Enable MCP + web search	Cross-reference claims against related literature, not just the paper itself.
Add an NLI-based second check	Combine cosine similarity with natural-language-inference to catch topically-similar-but-non-supporting evidence.
Feedback-driven calibration	Use corrected groundedness flags to recalibrate the threshold per domain over time.
Benchmark against baseline	Extend baseline_agent.py comparisons into a small benchmark with human-labeled ground truth.

Conclusion

The AI Research Paper Skeptic Agent turns paper reading from passive summarization into active, evidence-checked review. By separating claim generation (Skeptic) from claim verification (Guardrail), and grounding every claim in retrieved passages, the system gives readers a structured starting point for critical evaluation — while keeping final judgment with the human reader.

4

collaborating agents

6

build layers

0.6

default grounding threshold

Thank you

Questions welcome.

Repository: Scholar_Research_Agent
Team: [Add team member names]